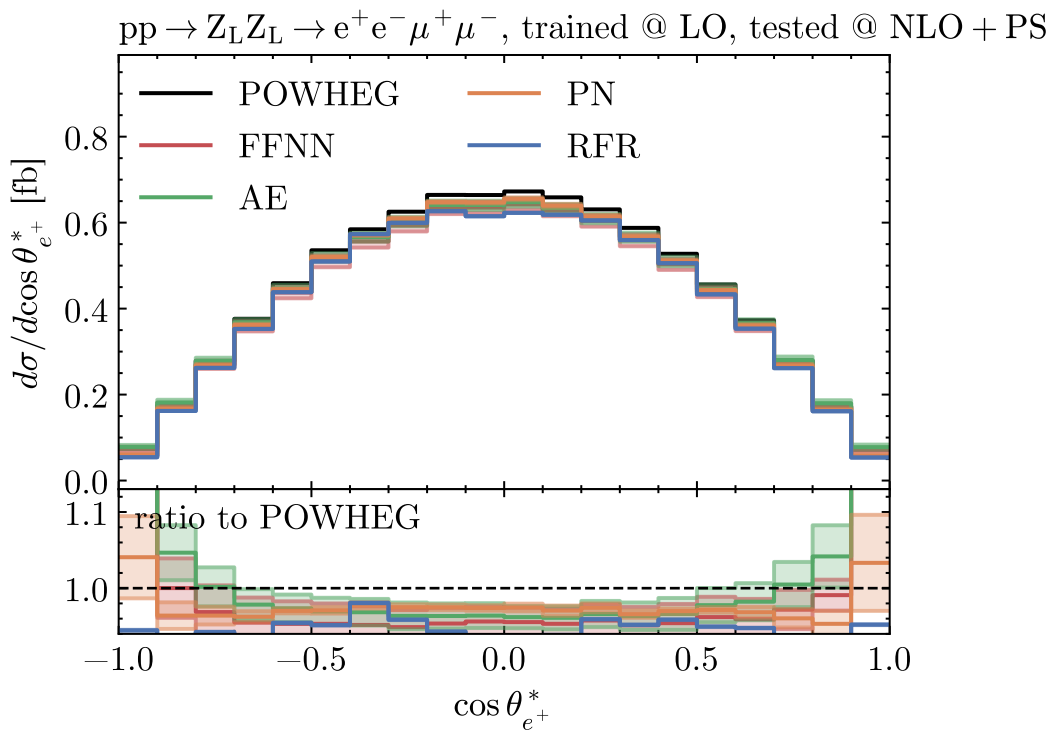
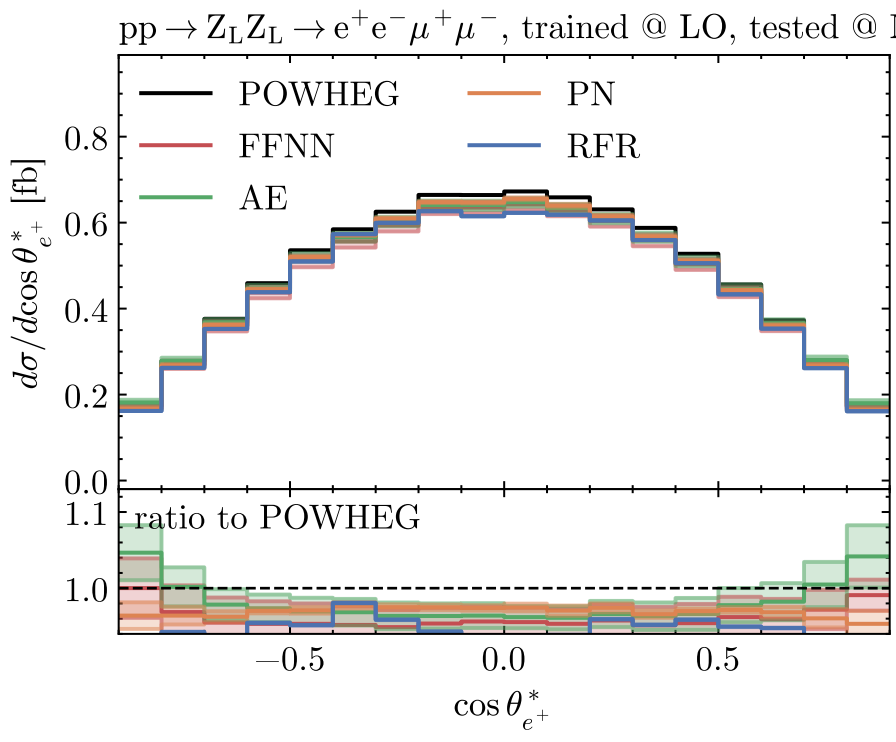
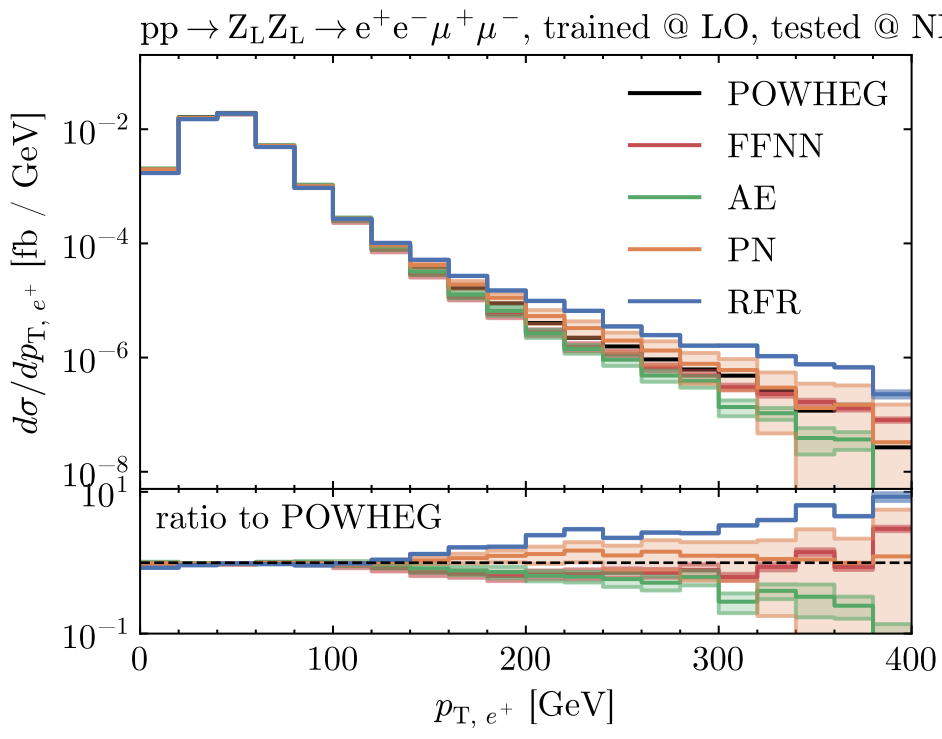
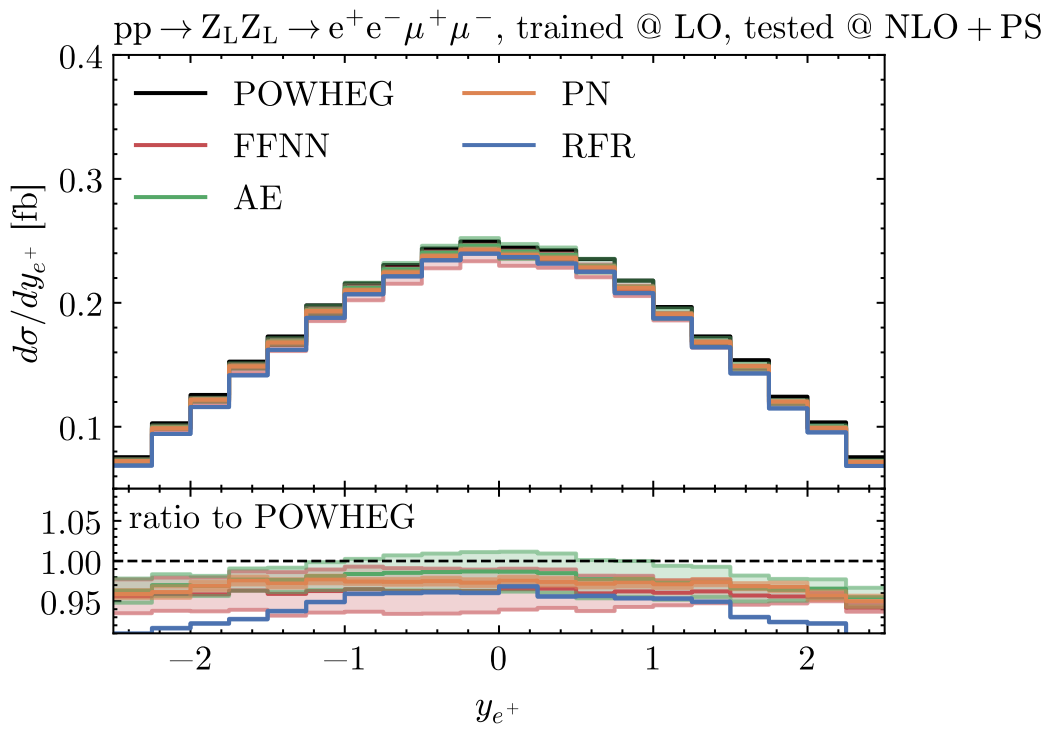
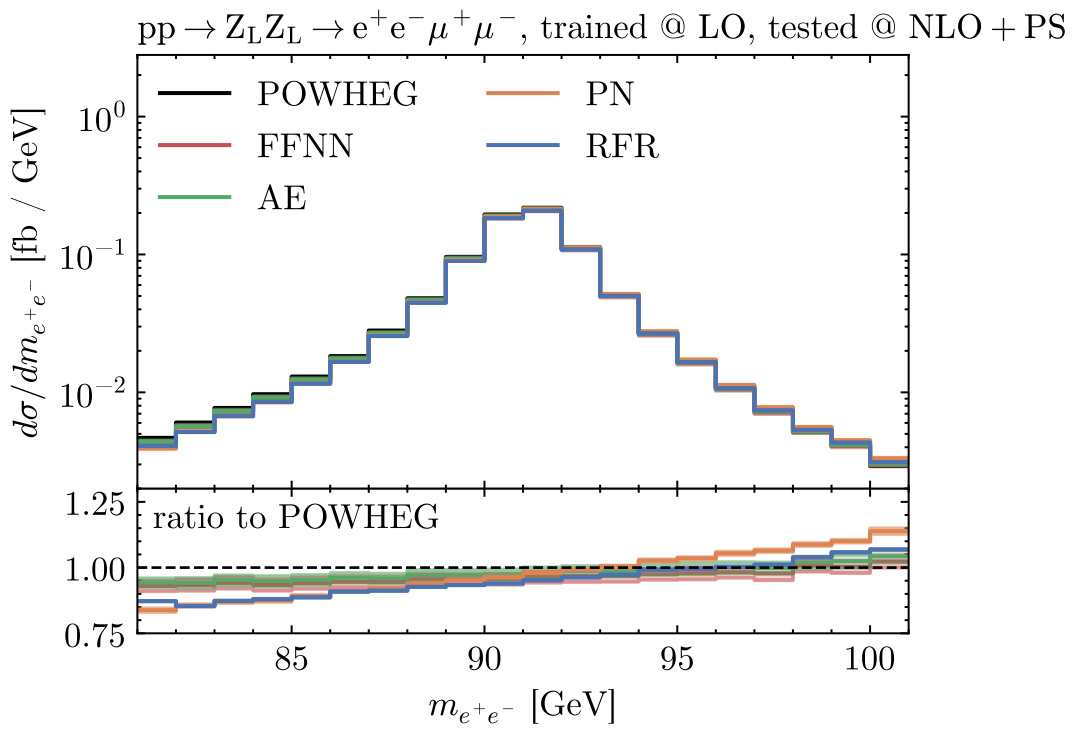
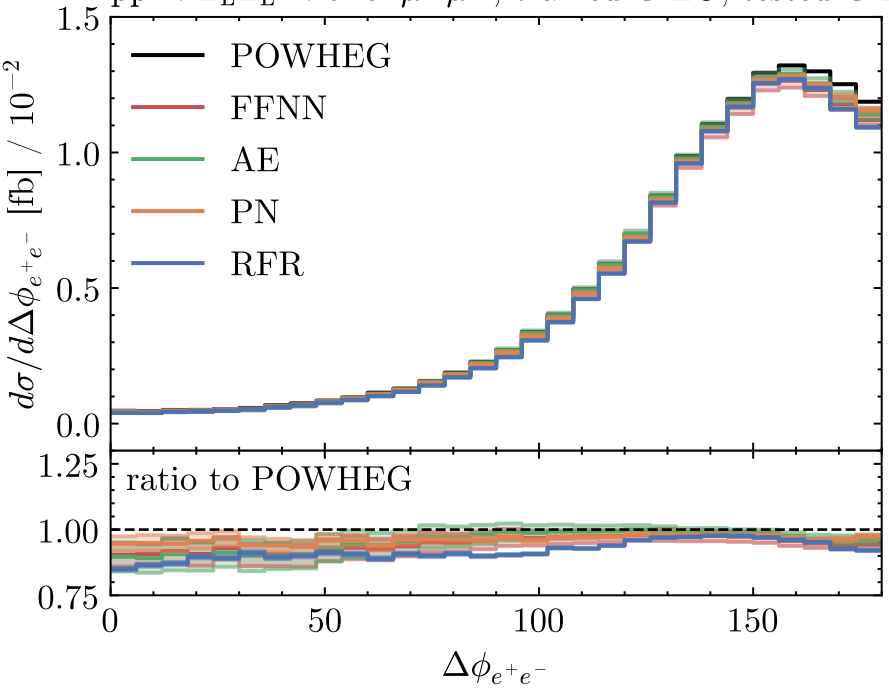

$\mathrm{pp} \to \mathrm{Z_L Z_L} \to \mathrm{e^+ e^- \mu^+ \mu^-}$, trained @ LO, tested @ NLO + PS
POWHEG
FFNN
AE
PN
RFR
$d\sigma / d\cos\theta^*_{e^+}$ [fb]
0.0
0.2
0.4
0.6
0.8
ratio to POWHEG
1.0
1.1
−1.0
−0.5
0.0
0.5
1.0
$\cos\theta^*_{e^+}$

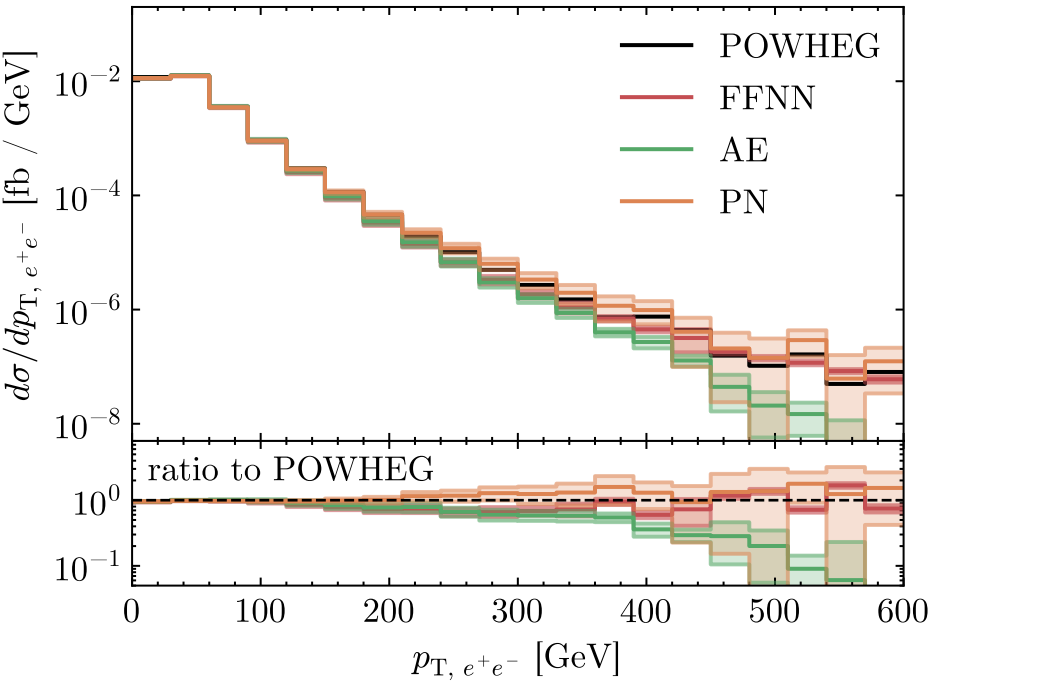

$pp \to Z_L Z_L \to e^+ e^- \mu^+ \mu^-$, trained @ LO, tested @ NLO + PS
POWHEG
FFNN
AE
PN
RFR
$d\sigma/d\cos\theta^*_{e^+}$ [fb]
0.0
0.2
0.4
0.6
0.8
ratio to POWHEG
1.0
1.1
−0.5
0.0
0.5
$\cos\theta^*_{e^+}$

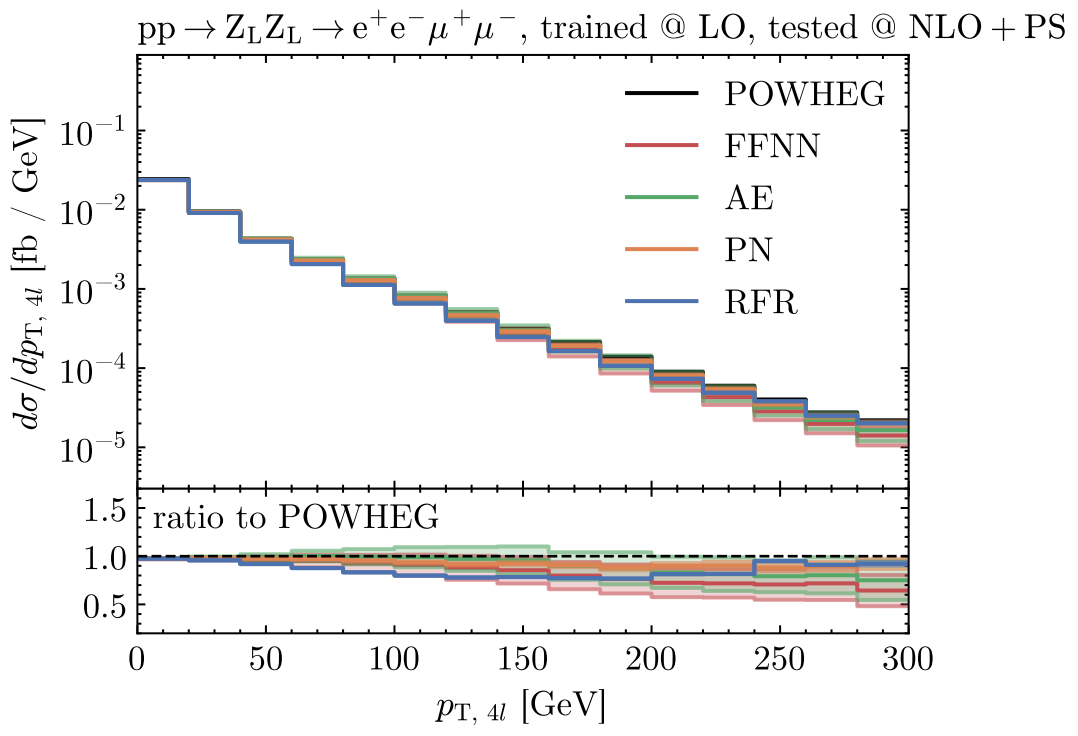

pp → Z_L Z_L → e⁺e⁻μ⁺μ⁻, trained @ LO, tested @ NLO + PS
POWHEG
FFNN
AE
PN
RFR
dσ/dp_T, e⁺ [fb / GeV]
ratio to POWHEG
p_T, e⁺ [GeV]
0
100
200
300
400

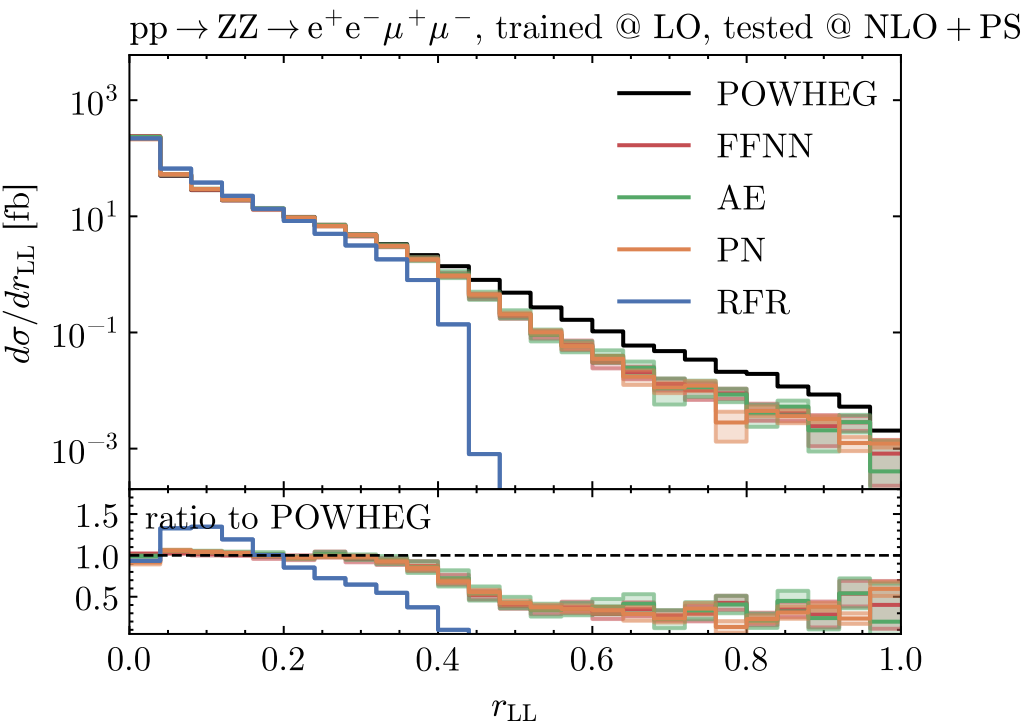

$\mathrm{pp} \to \mathrm{Z_L Z_L} \to \mathrm{e^+ e^- \mu^+ \mu^-}$, trained @ LO, tested @ NLO + PS
POWHEG
FFNN
AE
PN
RFR
$d\sigma/dy_{e^+}$ [fb]
0.4
0.3
0.2
0.1
ratio to POWHEG
1.05
1.00
0.95
−2
−1
0
1
2
$y_{e^+}$

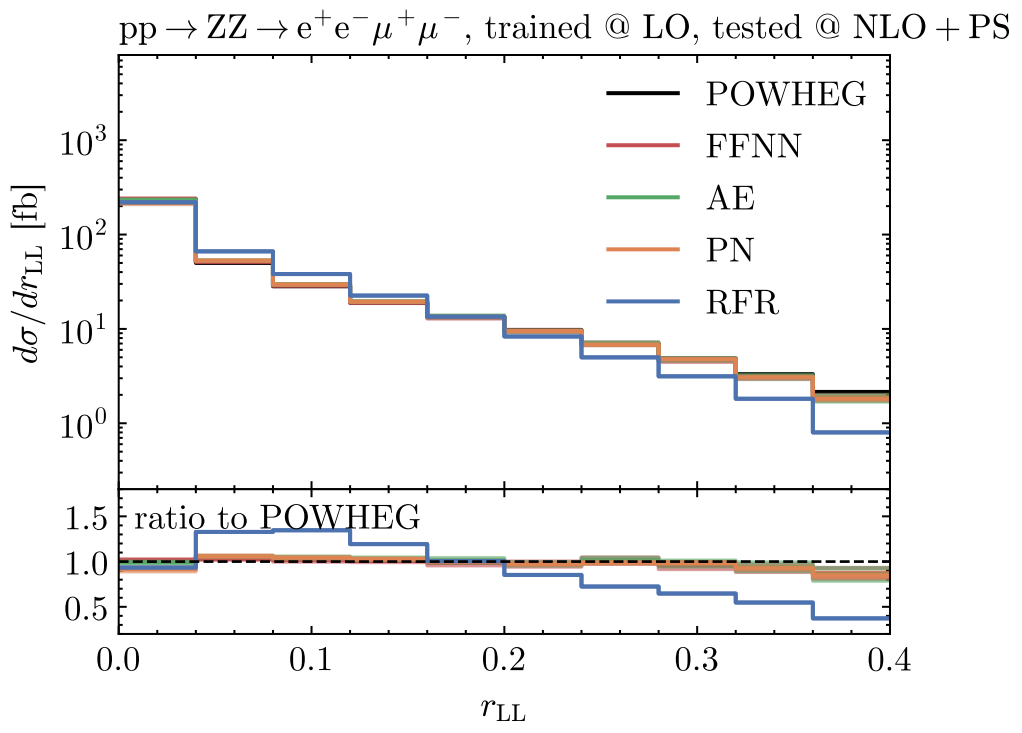

$\mathrm{pp} \to \mathrm{Z_L Z_L} \to \mathrm{e^+ e^- \mu^+ \mu^-}$, trained @ LO, tested @ NLO + PS
POWHEG
FFNN
AE
PN
RFR
$d\sigma/dm_{e^+e^-}$ [fb / GeV]
$10^0$
$10^{-1}$
$10^{-2}$
ratio to POWHEG
1.25
1.00
0.75
85
90
95
100
$m_{e^+e^-}$ [GeV]


pp → $Z_L Z_L$ → $e^+e^-\mu^+\mu^-$, trained @ LO, tested @ NLO + PS
POWHEG
FFNN
AE
PN
RFR
$d\sigma/d\Delta\phi_{e^+e^-}$ [fb] / $10^{-2}$
1.5
1.0
0.5
0.0
ratio to POWHEG
1.25
1.00
0.75
0
50
100
150
$\Delta\phi_{e^+e^-}$


$\mathrm{pp} \to \mathrm{Z_L Z_L} \to \mathrm{e^+ e^- \mu^+ \mu^-}$, trained @ LO, tested @ NLO + PS
POWHEG
FFNN
AE
PN
$d\sigma/dp_{\mathrm{T},\,e^+e^-}$ [fb / GeV]
$10^{-2}$
$10^{-4}$
$10^{-6}$
$10^{-8}$
ratio to POWHEG
$10^{0}$
$10^{-1}$
0
100
200
300
400
500
600
$p_{\mathrm{T},\,e^+e^-}$ [GeV]


pp → $Z_L Z_L$ → $e^+e^-\mu^+\mu^-$, trained @ LO, tested @ NLO + PS
POWHEG
FFNN
AE
PN
RFR
$d\sigma/dp_{T,\,4l}$ [fb / GeV]
$10^{-1}$
$10^{-2}$
$10^{-3}$
$10^{-4}$
$10^{-5}$
ratio to POWHEG
1.5
1.0
0.5
0
50
100
150
200
250
300
$p_{T,\,4l}$ [GeV]


pp → ZZ → e⁺e⁻μ⁺μ⁻, trained @ LO, tested @ NLO + PS
POWHEG
FFNN
AE
PN
RFR
dσ/dr_LL [fb]
ratio to POWHEG
r_LL


$pp \to ZZ \to e^+e^-\mu^+\mu^-$, trained @ LO, tested @ NLO + PS
POWHEG
FFNN
AE
PN
RFR
$d\sigma/dr_{LL}$ [fb]
$10^3$
$10^2$
$10^1$
$10^0$
ratio to POWHEG
1.5
1.0
0.5
0.0
0.1
0.2
0.3
0.4
$r_{LL}$